

Leveraging Open-Source Intelligence for Automated BGP Incident Attribution via LLM-based AI Agents

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A Motivating Example

A BGP hijacking incident that appeared ‘technically anomalous’ was later found to be a benign misconfiguration, after we manually gathered open information and reasoned it through:

- 1. AS17894’s parent organization, **Innovate Communications**, is a subsidiary of **Globe Telecom**
- 2. AS3758’s parent organization, **SingNet**, is operated by **SingTel**
- 3. **SingTel** is the principal shareholder of **Globe Telecom**

Internet and TV [edit]

- **SingNet** – provision of internet access and pay television services
- Optus Broadband Pty Limited – provision of high speed residential internet service
- Optus Vision Pty Limited – provision of interactive television service
- Optus Internet Pty Limited – provision of internet services to retail customers
- Vividwireless Group Limited – provision of wireless broadband services

Mobile company	Country	Stake ^[49]	Market Position ^[49]	Country Mobile Share Data							
				as of 31 March 2023 ^[50]	as of 31 March 2022 ^[51]	as of 31 March 2021 ^[52]	as of 31 March 2020 ^[53]	as of 31 March 2019 ^[49]	as of 30 June 2018 ^[54]	as of 31 March 2017 ^[55]	as of 31 March 2016 ^[56]
Advanced Info Service	Thailand	23%	No. 1	47.8%	46%	46%	45.2%	45%	45%	45%	47%
Bharti Airtel	India	32%	No. 2	32.4%	31.6%	29.8%	28.4%	28%	31%	23%	24%
Globe Telecom	Philippines	47%	No. 1	56.4%	55.4%	52.6%	55%	57%	52%	48%	46%

Wikipedia snippet of “SingTel”

Subsidiaries [edit]

- 917Ventures – 100% ownership
 - AdSpark Inc.
 - Global Telehealth, Inc. (KonsultaMD)
 - Inquiro – 49% ownership
 - Mynt (formerly Globe Fintech Innovations, Inc.) – 45% ownership; co-owned with Ayala Group and **Ant Financial**
 - Fuse Lending – mobile financial solutions provider, marketed under the GLoan and GGives brands
 - G-Xchange (GXI) – mobile payment and remittance service, marketed under the **GCash** brand
 - Rappit (in partnership with **Puregold**) – 50% ownership. Formerly PureGo.
 - Rush – 49% ownership
- Asticom Technology, Inc. – 100% ownership
- **Bayan Telecommunications**, Inc. (BayanTel) – 98.57% ownership
- Flipside Publishing Services, Inc. (FPSI) – 40% ownership
- GTI Business Holdings (GTI) – 100% ownership
- **Innovate Communications**, Inc. (Innovate) – 100% ownership
- Kickstart Ventures, Inc. (Kickstart) – 100% ownership
- Kroma
- Yondu (formerly Entertainment Gateway Group Corp.) – 100% ownership

Wikipedia snippet of “Globe Telecom”

First of all, why incident intent matters?

- Not every reported BGP incident is an attack; many are just benign mistakes
- Harmless errors can be many, and real attacks are buried
- Knowing intent lets us focus on the incidents that truly matter
- Operators rarely react to benign cases (from our experience), but move fast on malicious threats

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Challenges on this?

- Hidden relationships often reveal whether an event is attack or mistake
- Example: victim holding a service for “hijacker” → likely misconfiguration
- Traditional routing data largely misses these relationships (even CAIDA’s AS business relationships)
- Clues live in external sources (affiliations, partnerships, history)
- Yet, information is vastly scattered and unstructured, hard to retrieve and reason over

Can LLM-based AI agents be the solution?

- Automate reasoning with an agentic LLM + RAG framework
- Decide what context (intelligence sources) is needed for each BGP event adaptively
- Gather info (via existing search engines?) from diverse open sources, beyond routing data
- Synthesize evidence to reveal hidden relationships (ownership, service ties)
- Distinguish attacks from mistakes, with human-readable reports & confidence scores

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Intelligence sources

- General knowledge: Wikipedia, official websites
- Business intelligence: Yahoo Finance, Reuters, Crunchbase
- Operational sharing: APNIC/RIPE/LACNIC blogs, MANRS, CDN blogs and mailing lists
- Cybersecurity: Spamhaus, AbuseIPDB, Scamalytics
- Social Signals: Twitter, Facebook, LinkedIn announcements

Discussions

- Who actually need it?
- How to control the cost?
- Scalability to large scale incident analysis?
- A fully automated agent, or a human-supervised workflow with LLMs?